Title: Library Management System

Problem

Many libraries still rely on manual processes for cataloguing books, tracking inventory, and managing borrowing records.

Solution

Developed a Python- and MySQL-based Library Management System that automates book management, user management, issue/return tracking, and personal note-taking.

Key Features

- Role-based login (Admin and Member)
- Book inventory management
- User account management
- Book issue and return tracking
- Search functionality
- Personal notes module
- Secure database interaction using parameterised SQL queries

Technologies Used

- Python
- MySQL
- MySQL Connector
- SQL

Key Learning Outcomes

- Database design
- Secure coding practices
- Software development lifecycle
- Problem-solving through technology

Future Enhancements

- Graphical user interface
- Password encryption
- Format validation (Email, Telephone, Year etc.)
- Barcode integration
- Recommendation engine

The project helped me understand how systems are designed. I realised that technology is not just about writing code. It is about understanding how people work, how information is organised and how systems can be designed to make processes more efficient. Working on this project strengthened my interest in exploring the connection between technology, society, and problem-solving.